

CS-301 Microprocessor Based System Design

Course Teacher: Dr.-Ing. Shehzad Hasan

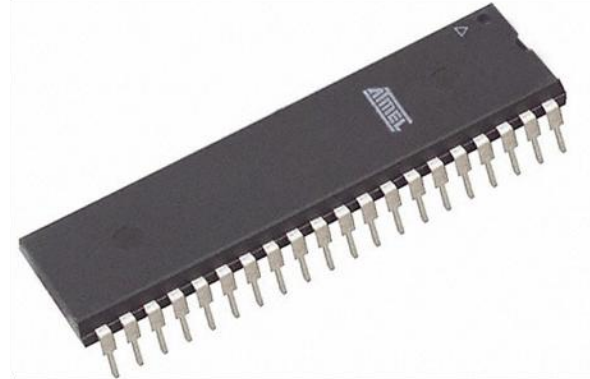
Lecture - 4

Processor Technology

- The architecture of the computation engine used to implement a system's desired functionality.
- The term *processor* is usually associated with programmable software processors, but here we consider nonprogrammable, digital systems as being processors too.



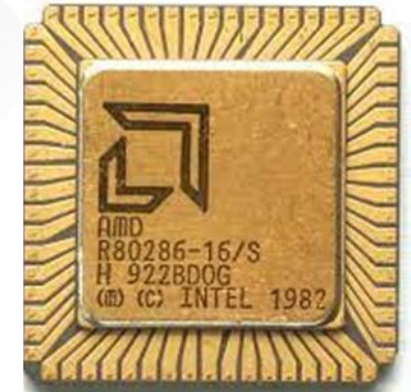
Digital Signal Processor



Microcontroller



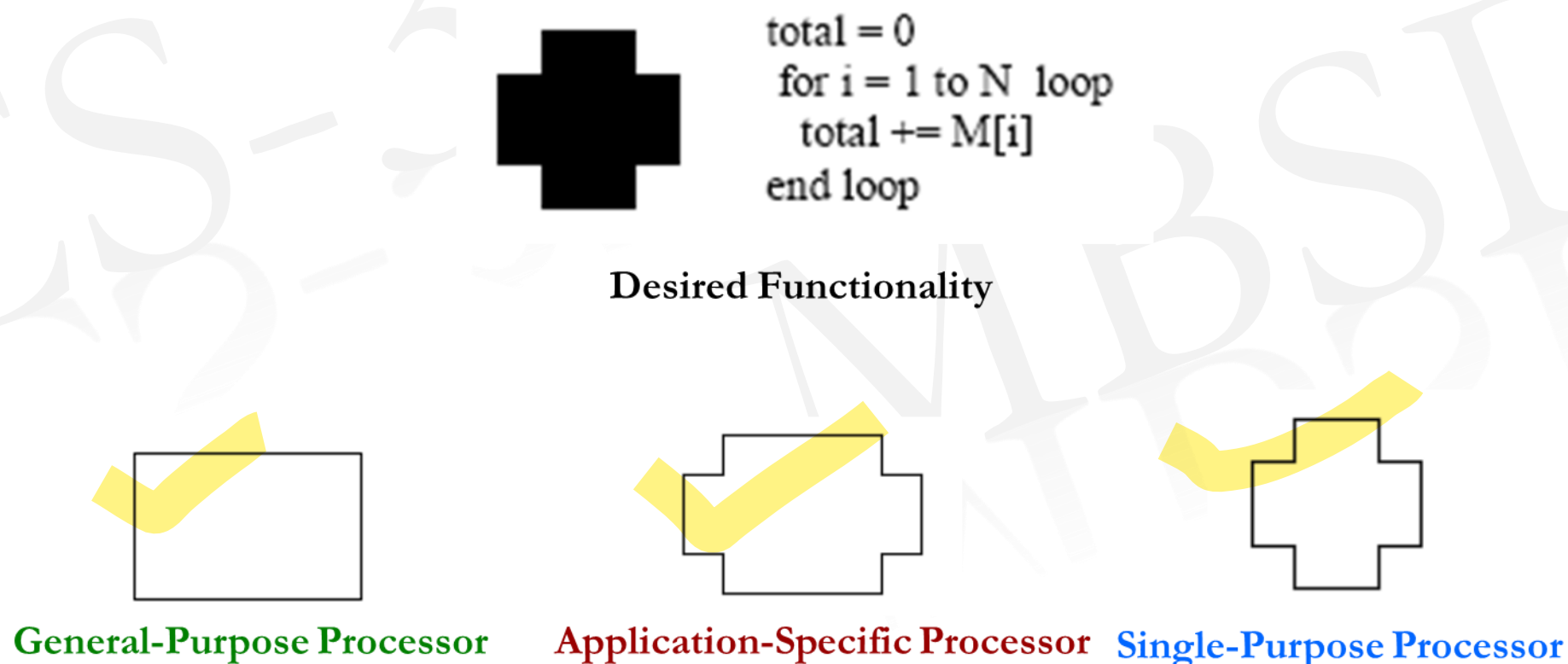
Encryption chip



Desktop Processor

Comparison of Processor Technologies

- Processors vary in their customization for the problem at hand



Processor Technology

- **General-Purpose Processors**

- Programmable device used in a variety of applications, also known as “microprocessor”

- **Features**

- **Program memory** It's where your device keeps the steps it needs to follow to run software.
- **General datapath with large register file and general ALU**

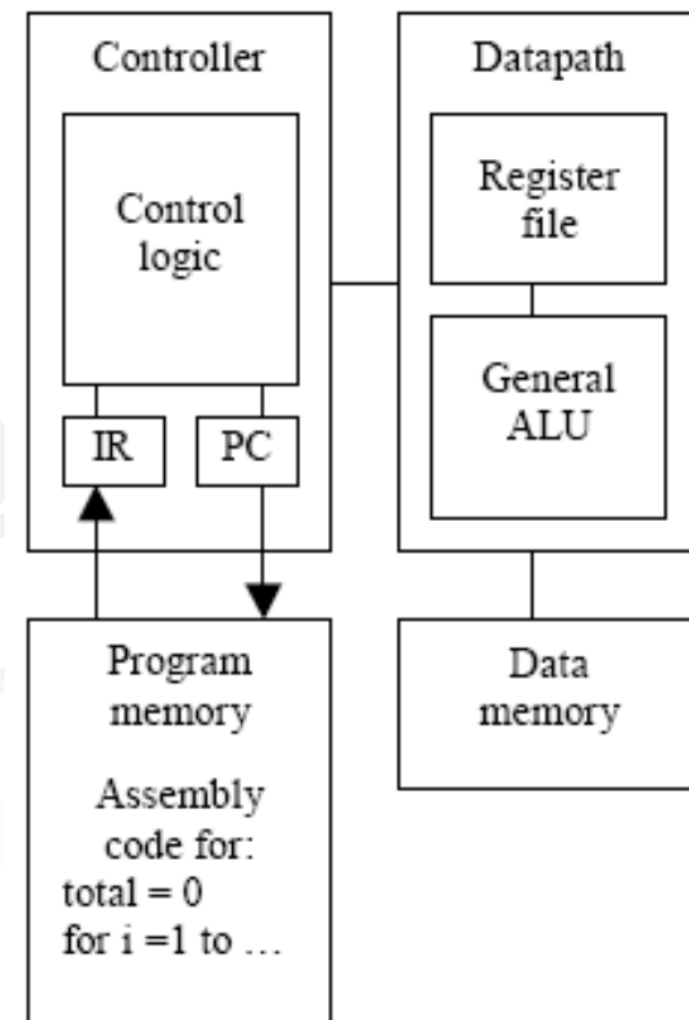
- **Focus is on software**

- **User benefits**

- Low time-to-market and NRE costs
- High flexibility
- Per product cost is low for small quantities
- Performance may be fast for computation intensive applications if a fast and expensive processor is used

- **Drawbacks**

- Increases size and power due to unnecessary hardware
- Per product cost may be relatively high for large quantities
- Performance may be slow for certain applications



Processor Technology

• Single-Purpose Processors

- Digital circuit designed to execute exactly one program a.k.a. coprocessor, accelerator or peripheral

– Features

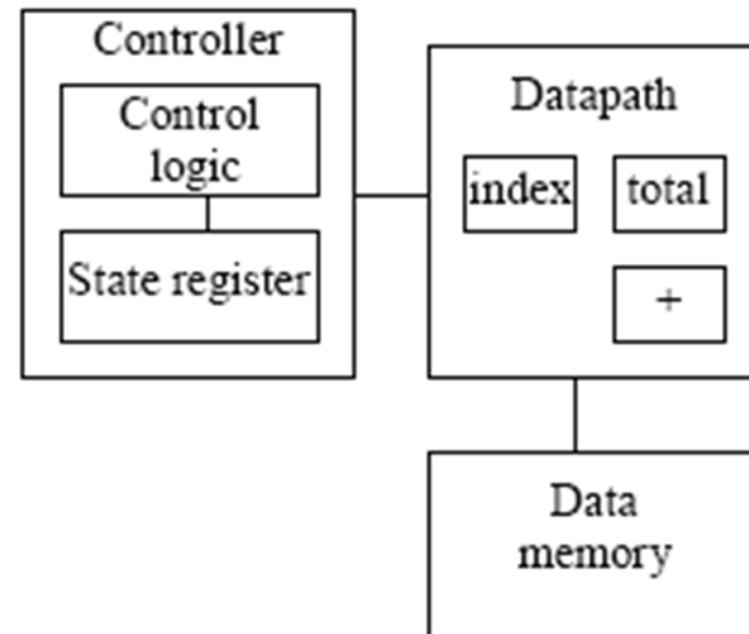
- Contains only the components needed to execute a single program (functionality)
- Program's instructions are hardwired into the control logic
- No program memory but is resented in gpp

– Benefits

- Performance is good, fast operations due to hardwired logic
- Low power
- Small size
- Per product cost is low for large quantities

– Drawbacks

- High design time and NRE costs
- Low flexibility
- Per product cost is high for small quantities



Processor Technology

- Single purpose processor outperforms software based processor by 2-3 orders of magnitude in terms of performance, power efficiency but require a lot more engineers to implement.
- Examples of single-purpose processors
 - I/O controllers (USB, I2C, SPI etc.)
 - Encryption/Decryption processor
 - SHA-256 algorithm (Blockchain and cryptocurrency)
 - Video Encoders/Decoders e.g. (H.264 MPEG, H.265 HEVC etc.)
 - Audio Synthesizers
 - Fixed Function FPUs (Floating Point Units)
 - Checksum Calculators/CRC generators
 - Noise suppression Filters
 - Signal Demodulators (QAM demodulators in TV set-top boxes)

Processor Technology

- Application-Specific Instruction-Set Processors or **Special Purpose Processors**

- Programmable processor optimized for a particular class of applications having common characteristics

- Compromise between general-purpose and single-purpose processors

- **Features**

- Program memory
 - Optimized datapath
 - Special functional units

- **Benefits**

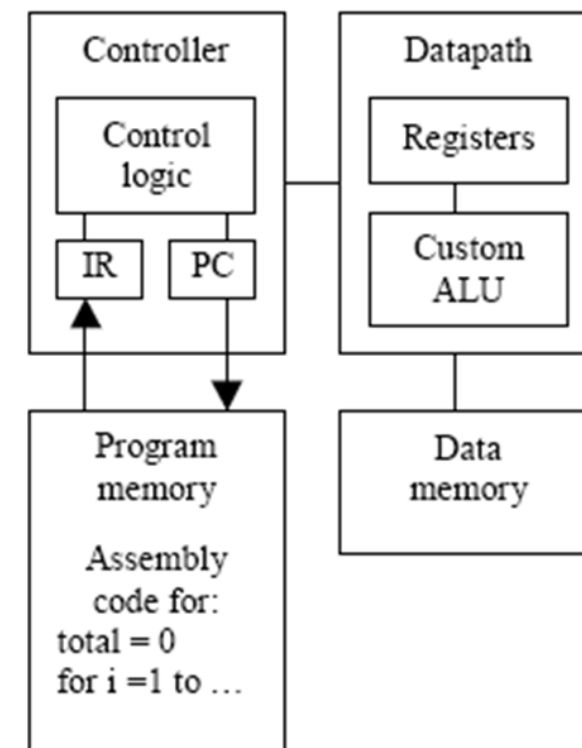
- Some flexibility, good performance, size and power

- **Drawbacks**

- High NRE cost to build the processor itself and to build a compiler if they don't already exist

- **Examples**

- **Microcontrollers** (Specialized for I/O operations)
 - **DSPs** (Digital Signal Processing applications)
 - **GPUs** (Specialized for graphics operations)
 - **PLCs** (Programmable Logic Controllers)



IC Technology

designing a digital circuit using basic logic gates

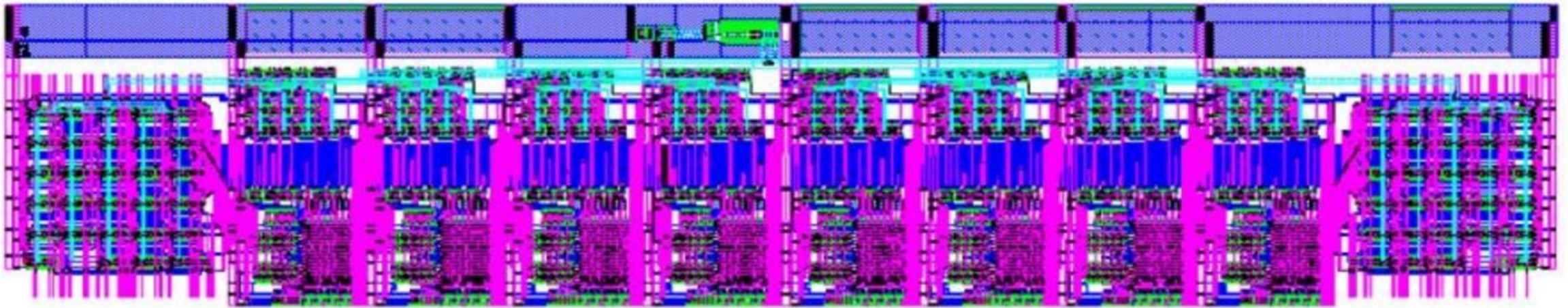
- The manner in which a digital (gate-level) implementation is mapped onto an IC
- Three Types of IC Technologies
 - Full-custom design
 - Semi-custom design / ASIC
 - Programmable Logic device/FPGA

IC Technology

Only way to design analog and ultra high speed devices

- **Full-custom design**

- Every transistor is designed and drawn by hand
- All layers are optimized for an embedded system's particular digital implementation
 - Placing transistors to minimize interconnection lengths
 - Sizing transistors to optimize signal transmissions and power
 - Routing wires among transistors



IC Technology

- Full-custom design

- Benefits

- Excellent performance, small size, low power

- Drawbacks

- High NRE cost
 - Long time-to-market

- Commonly used only in high-volume or extremely performance-critical applications

- High-Performance Computing (HPC) and AI Accelerators

- **Tensor Processing Units (TPUs)** or specialized accelerators for machine learning often use full-custom designs to optimize for specific workloads like matrix operations, convolutional layers, and data flow control

- Power-Critical Applications

- IoT sensors with power budgets in the microwatt range
 - Implantable medical devices

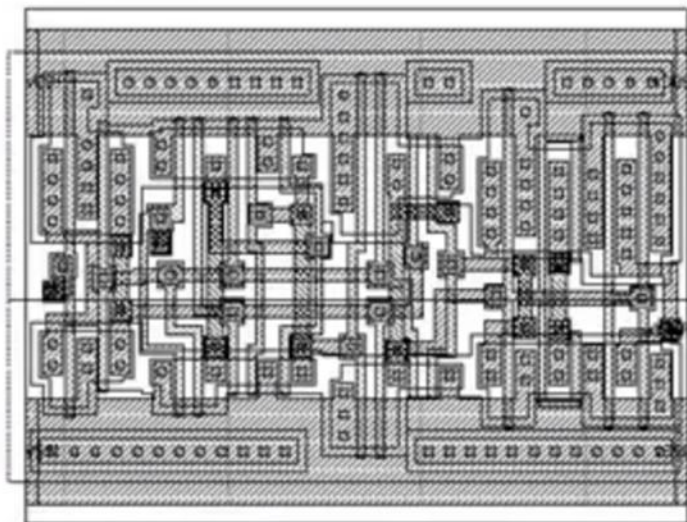
- Analog and Mixed-Signal Design (RF Front ends for 5G networks)

- Memory Design (High speed cache memory)

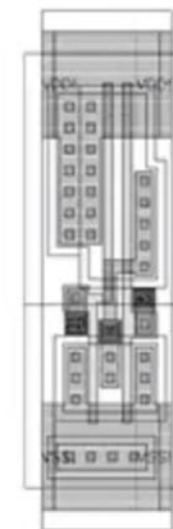
IC Technology

- **Semi-Custom/ASIC (Application Specific Integrated Circuit)**
 - Standard-cell design (Cell Based IC or semi-custom)
 - Standard cells are custom designed and then inserted into a library
 - These cells are then used in the design by being placed in rows and wired together using 'place and route' CAD tools
 - RTL, Verification, Logic synthesis on standard-cells → netlist, Placement, Routing, Photomasks

D Flip-Flop



NOR gate



IC Technology

- Semi-Custom/ASIC (Application Specific Integrated Circuit)
 - Benefits
 - Good performance, good size
 - less NRE cost than a full-custom implementation
 - Drawbacks
 - Full set of masks (around 22) still required
 - Still require weeks or months to develop

IC Technology

- Programmable Logic devices/FPGA

- All layers already exist

- Designers can purchase an IC

- Connections on the IC are either created or destroyed to implement desired functionality or there can be a look-up table (LUT/ memory based) approach to implement the functionality

- Field-Programmable Gate Arrays (FPGA) are very popular

- Benefits

- No custom mask layers so final design implementation is few hours instead of few weeks

- Low NRE costs, almost instant IC availability,

- Drawbacks

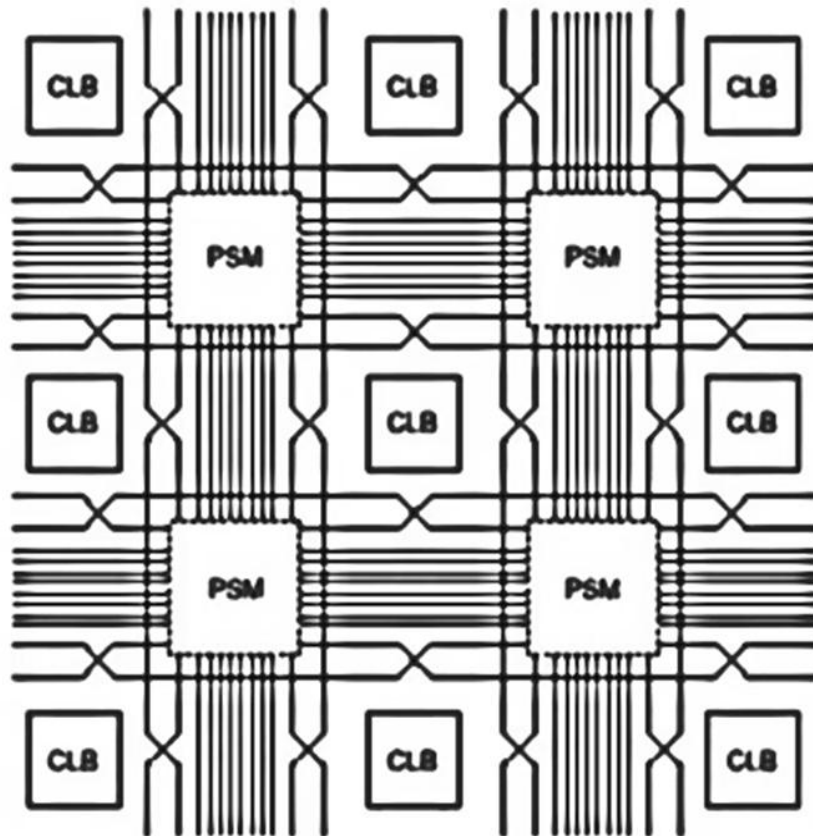
- Bigger, expensive, power hungry and slower than ASIC

- High per unit cost

IC Technology

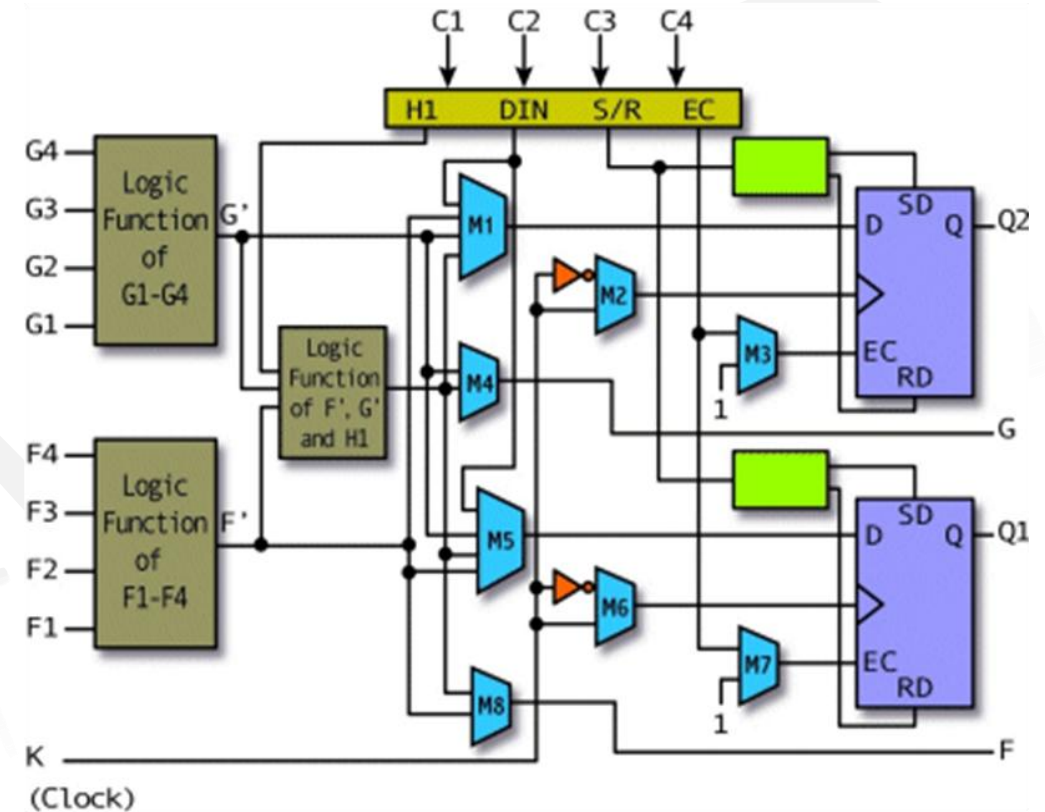
- Programmable Logic devices/FPGA

Programmable interconnect array



PSM –
Programmable
Switch Matrix

Configurable Logic Block (CLB)



Mapping Processor Technology to IC Technology

Any processor technology can be mapped to any IC technology.

